

Muhammad Al-Aasar

Junior AI Engineer & Data Analyst

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ABOUT

Junior AI Engineer & Data Analyst with a Computer Science degree from Tanta University (GPA 3.1/4). Hands-on experience across Machine Learning, Deep Learning, Generative AI, and Agentic/RAG systems, backed by a strong data-analysis foundation in Python, SQL, and BI tooling. Built and shipped end-to-end projects spanning computer vision, NLP, multi-agent systems, workflow automation, and production-style RAG pipelines. Kaggle Expert, and led a 4-person team to an A+ graduation project.

EDUCATION

Bachelor's Degree in Computer Science (GPA: 3.1/4)

Sep 2020 - Sep 2024

Faculty of Computers and Information, Tanta University

PROFESSIONAL EXPERIENCE

Think Company — Generative AI Intern

Jul 2025 - Nov 2025 | Cairo, Egypt

- Worked on diffusion models and AI tools, contributing to the company's generative AI initiatives.
- Experimented with prompt-based generation techniques and fine-tuning approaches (LoRA, PEFT) to adapt models for specific use cases.
- Collaborated with the AI team to evaluate model outputs and improve generation quality.

Instant Academy — AI & Data Analyst Practical Training

15 Sep 2025 - 15 Oct 2025 | Cairo, Egypt

- Worked on Financial Transactions Analysis, Brain Tumor Classification & Segmentation, Car License Plate Detection, Sentiment Analysis, and Text Summarizer & Question Answering projects.
- Applied end-to-end data analysis and deep learning workflows, from data cleaning and preprocessing to model training and evaluation.
- Gained hands-on experience across computer vision, NLP, and business intelligence tasks in a structured practical-training environment.

Zewail University — Machine Learning Trainee

Aug 2023 - Nov 2023 | Cairo, Egypt

- Trained and evaluated Machine Learning models on real-world datasets.
- Worked on data preprocessing, feature selection, and model optimization.
- Built a solid foundation in core ML workflows under trainer guidance, ahead of progressing to deep learning and Generative AI.

South Delta Electricity Distribution Company — Data Analyst Intern

Aug 2023 - Sep 2023 | Tanta, Egypt

- Analyzed and cleaned data using Python (Pandas, NumPy).
- Created Power BI dashboards and optimized queries with SQL.
- Supported reporting processes by turning raw operational data into clear, actionable dashboards for stakeholders.

PROJECTS

Graduation Project: AI Integration in an E-Learning Platform — Grade: A+ | Team Leader

- Led a team of 4 members, supervising project workflow and ensuring seamless integration of AI components into a web and mobile-based e-learning platform.
- Developed two AI models to enhance platform functionality, focusing on computer vision and NLP.
- Implemented an AI-powered identity recognition system using OpenCV & OCR for automated ID verification.
- Created an NLP-based model to detect offensive language in course interactions, achieving 95%+ accuracy with TF-IDF & word embeddings.

Graph RAG — Egyptian Premier League Knowledge Assistant

- Built a Knowledge Graph covering the Egyptian Premier League (clubs, players, transfer history, match results, club trophies), sourcing data from Wikidata and Wikipedia and storing it in Neo4j.
- Connected the graph to an LLM via LangChain's GraphCypherQChain, translating natural-language questions (Arabic or English) into Cypher queries and returning natural-language answers instead of raw IDs.
- Implemented an entity-resolution layer using BAAI/bge-m3 embeddings to match player/club names across spelling variations before query execution.

- Built a FastAPI backend with a lightweight Arabic chat interface, fully containerized with Docker for one-command setup; open-sourced the full codebase.

WE (Telecom Egypt) — RAG Chat System

- Built a web crawler to scrape the official te.eg website (respecting robots.txt), then split the scraped text into chunks using a RecursiveCharacterTextSplitter and stored them in PostgreSQL.
- Generated embeddings using the BAAI/bge-m3 model and stored them in PostgreSQL with the pgvector extension.
- Implemented hybrid retrieval combining dense (semantic, pgvector) and sparse (lexical, Postgres full-text) search, fused using Reciprocal Rank Fusion (RRF).
- Integrated the Groq API (Llama / GPT-OSS models) for grounded generation with a minimum relevance-score threshold and prompt-injection guardrails.
- Implemented checkpointing of chat sessions and messages in PostgreSQL, with retrieval or deletion of any past conversation.

CNN Project: Brain Tumor Detection & Segmentation App

- Developed and trained a U-Net architecture from scratch for brain tumor segmentation using CNNs with Batch Normalization and Dropout.
- Built end-to-end preprocessing pipelines for MRI scans, including resizing, normalization, and multichannel FLAIR handling.
- Compiled the model with Adam optimizer and a custom Dice Loss function to address class imbalance; evaluated via Mean IoU.
- Designed a user-friendly interface for medical professionals, comparing original MRI scans against predicted tumor masks.

Data Squad — Multi-Agent Data Analysis System

- Architected a multi-agent pipeline (Supervisor, SQL, Cleaner, Reporter agents) using LangGraph, FastAPI, and Groq (Llama 3.3 70B) to answer natural-language questions in Arabic/English against a SQL Server database.
- Implemented a human-in-the-loop checkpoint where LLM-generated pandas cleaning code is reviewed and approved before execution in a restricted sandbox.
- Built a SQL Agent that reads the live database schema and anchors date-relative queries to the most recent date in the data.
- Integrated MCP servers for SQL access, code sandboxing, and web-search enrichment (Tavily); delivered a live pipeline-visualization frontend with PDF export.

Planing_Trip AI — Multi-Agent Travel Planner

- Built a multi-agent travel planning system with LangGraph and FastAPI, coordinating Supervisor, Flight, Hotel, Weather, Budget, Itinerary, and Final agents.
- Integrated external tools via MCP, including AviationStack for flight data, Tavily for hotel search, and a custom OpenWeather-based MCP server.
- Designed a human-in-the-loop approval step for users to review, approve, or revise the draft itinerary.
- Persisted per-session conversation state using PostgreSQL (PostgresSaver) with LangGraph; served through a Jinja2-rendered web UI.

E-Commerce Automation Pipeline & Customer-Service Agent — n8n

- Designed an end-to-end n8n automation replacing a fully manual product-photography and posting workflow for a Facebook-based clothing store.
- Built a Google Drive-triggered pipeline that generates photorealistic AI model photos across 6 poses from two source images, using image-generation APIs with async job polling, retry logic, and rate-limit handling.
- Automated brand-logo watermarking, AI-generated captions, and direct publishing to Facebook via the Graph API, with a human-in-the-loop approval step via Telegram before any post goes live.
- Built a Facebook Messenger customer-service Agent with multi-model fallback (OpenAI, Google Gemini), handling product inquiries, voice-message transcription, and order booking against a live Google Sheets database.

CV Information Extraction — LoRA Fine-Tuning

- Fine-tuned Qwen2.5-1.5B-Instruct with LoRA to extract structured JSON (education, experience, certifications) from raw CV/resume text.
- Built an SFT dataset using a larger teacher model (Groq) to auto-label raw CV text with structured JSON extractions.
- Trained a rank-64 LoRA adapter with LLaMA-Factory, evaluated base vs. fine-tuned outputs, served with vLLM (--enable-lora), and load-tested with Locust.

Data Analysis: Financial Transactions Power BI Dashboard

- Cleaned and transformed raw transaction data using Power Query; designed a data model linking transactions, users, and cards for cross-filter analysis.
- Created custom DAX dimensions and integrated dynamic visuals, including custom maps and image URLs, for engaging storytelling.

Data Analysis: BMW Vehicle Sales Analysis

- Conducted EDA on a real-world vehicle dataset with missing values; engineered features to infer and fill missing attributes.
- Visualized global vehicle trends to analyze manufacturer distribution and body-type popularity.

COURSES

Digital Egypt Pioneers Initiative (DEPI) — Generative AI Track

Nov 2025 - Jul 2026 | Online

- Completed a 200-hour advanced AI program spanning ML/DL fundamentals, Generative AI, RAG, Advanced Prompt Engineering, and Agentic/Multi-Agent AI.
- Built Generative AI models (VAEs, GANs, Diffusion Models) and applied fine-tuning techniques such as LoRA and PEFT.
- Implemented RAG pipelines with advanced retrieval techniques (fusion, re-ranking) and memory-augmented context retention.
- Designed and deployed autonomous Agentic AI and Multi-Agent systems covering tool use, planning, memory, and agent coordination protocols.

Instant Academy — AI & Data Analyst Course

15 Oct 2024 - 15 Jul 2025 | Cairo, Egypt

- Analyzed and visualized data using Python (NumPy, Pandas) and Power BI, and managed relational databases with SQL.
- Built supervised & unsupervised ML models (Scikit-learn) and deep learning models (ANN, CNN) using TensorFlow and Keras.
- Applied computer vision (OpenCV, YOLO) and NLP (RNN, LSTM, Transformers) techniques, deploying ML apps with Streamlit.

SKILLS

- Data cleaning and preprocessing with NumPy, Pandas
- Deep Learning: FFNN, CNN, RNN, LSTM, GRU, Transformers
- Generative AI: Diffusion Models, Prompt-based Generation, Fine-tuning (LoRA, PEFT)
- Machine Learning: supervised and unsupervised algorithms
- Agentic AI: Autonomous Agent Architecture, Planning & Memory Systems
- Multi-Agent Systems: Communication, Coordination & Cooperation, Simulation
- Retrieval-Augmented Generation (RAG) & Graph RAG (Neo4j)
- Workflow Automation & Orchestration: n8n (Agentic Workflows, API Integration, Multi-step Automation Pipelines)
- Advanced Prompting & Reasoning: Chain-of-Thought, Tool Use & Function Calling, Context Optimization
- Data Visualization: Matplotlib, Seaborn, Power BI
- Business Intelligence (BI), Dashboard Development, Reporting
- SQL, Database Management, and Excel

LANGUAGES

- Arabic: Native
- English: Professional Proficiency

KEY ACHIEVEMENTS

- 3rd Place Winner in Instant Academy Analysis competition.
- Ranked as a Kaggle Expert with multiple high-quality notebooks and end-to-end deep learning projects.